CS-GY 6313
Information Visualization
Week 1: Introduction and Logistics

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This Course is about:

- A *bottom-up* tutorial on how a human-computer system performs: visual information *generation/consumption/perception/application,*
- *and your own hands-on creation!*

And Not About

- Data science/mining/…
- High-level “seminar” or “overview” – you will need to program ● How to draw “charts”
- Web development
- Data processing/statistics (excel, R, etc.)

Please:

- Interrupt me anytime
- Ask technical/engineering questions to me/TA
- Discuss with peers
- Use Version control your code (we’ll cover this today)
- Request extension/absence with a reason

We do not accept

- teamwork – independent work in the whole course
- copy code/text from online/others/any resource – we will perform plagiarism check: you **can** acknowledge and cite how you got guidance
- **F** if you do any of these

We will cover

- Mathematical and programming foundations ● Color theories and visual perception
- 2D visualization: spatial/temporal/network data
- 3D visualization: projections and case studies
- Modern topics: machine learning, etc.
- Guest lectures

Prerequisites

- Mathematics
- Basic Linear algebra
- Programming
  - **Python** (highlypreferred with Matplotlib) OR Javascript (if you have to)
- Academic Writing
- Latex

Grading Overview

- 15% x 4 Assignments (mini-projects)
- 10% Survey Article + Project Proposal
- 30% Final Project

Breakdown & Assignments

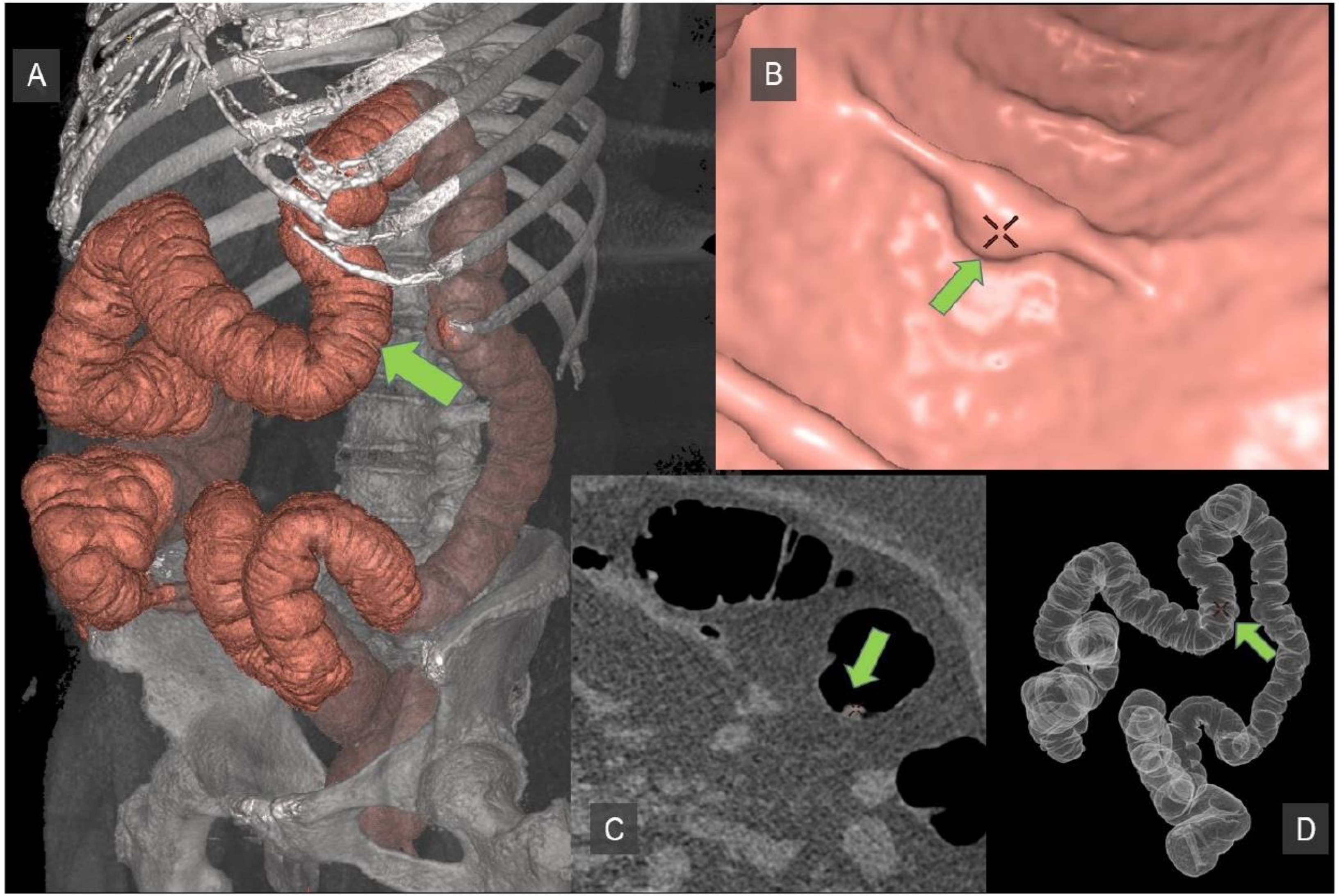
- Assignment 1-3: 2D visualization
- Assignment 4: 3D visualization
- Survey and final project – your choice! But do it throughout the course

Visualization

Any technique for creating **images/diagrams/animation** to communicate a message

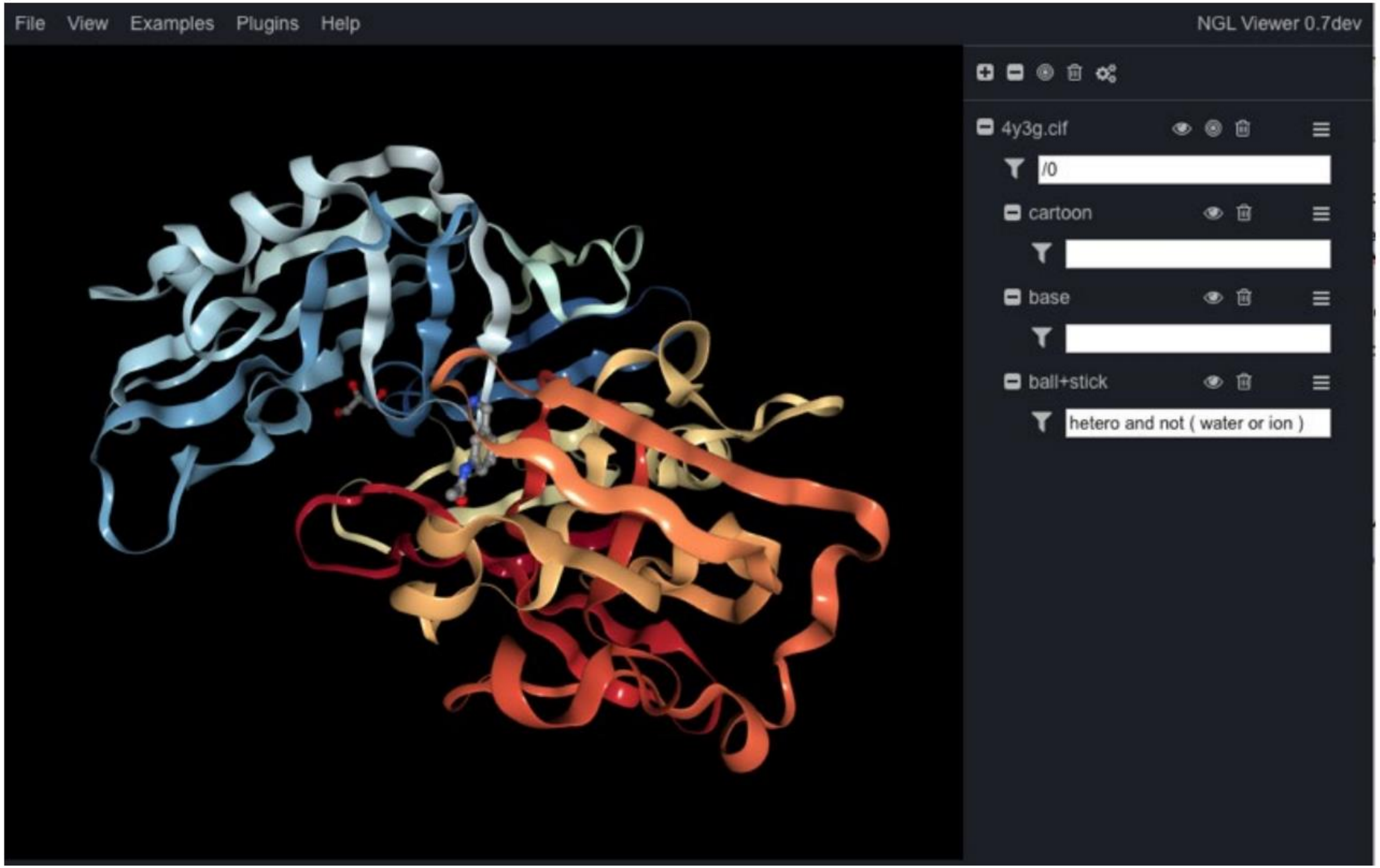
- Basic color and graphics theories
- Scientific visualization
- Information visualization
- Interfaces

Application of Visualization: Medical

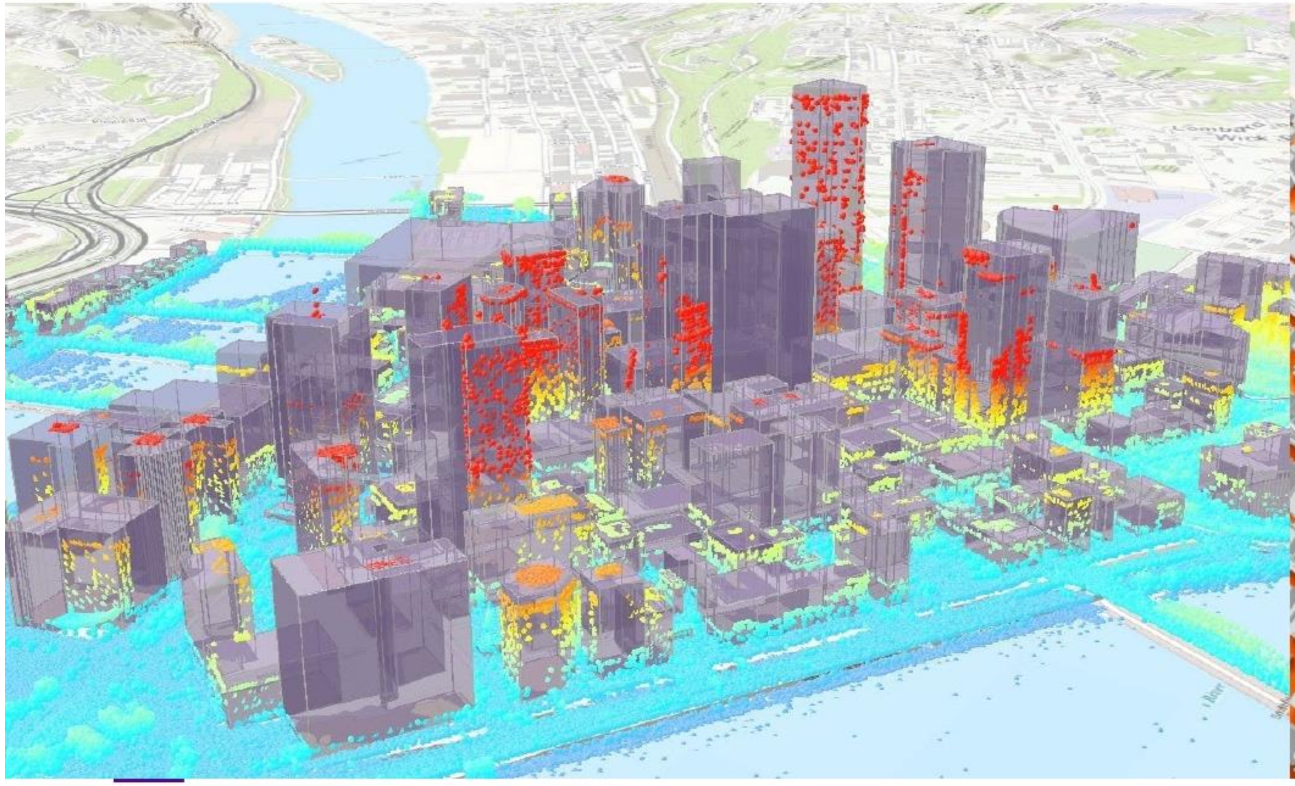


Application of Visualization: Finance

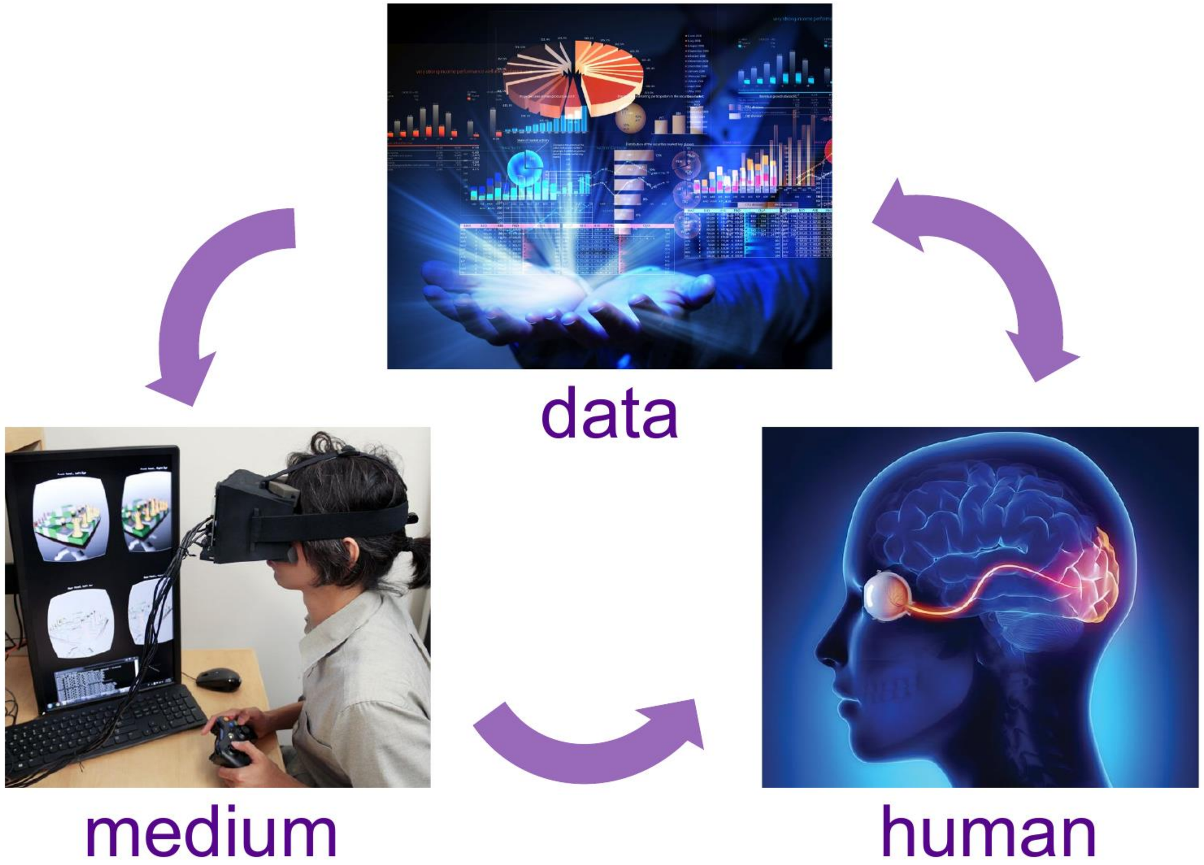
Application of Visualization: Scientific Discover



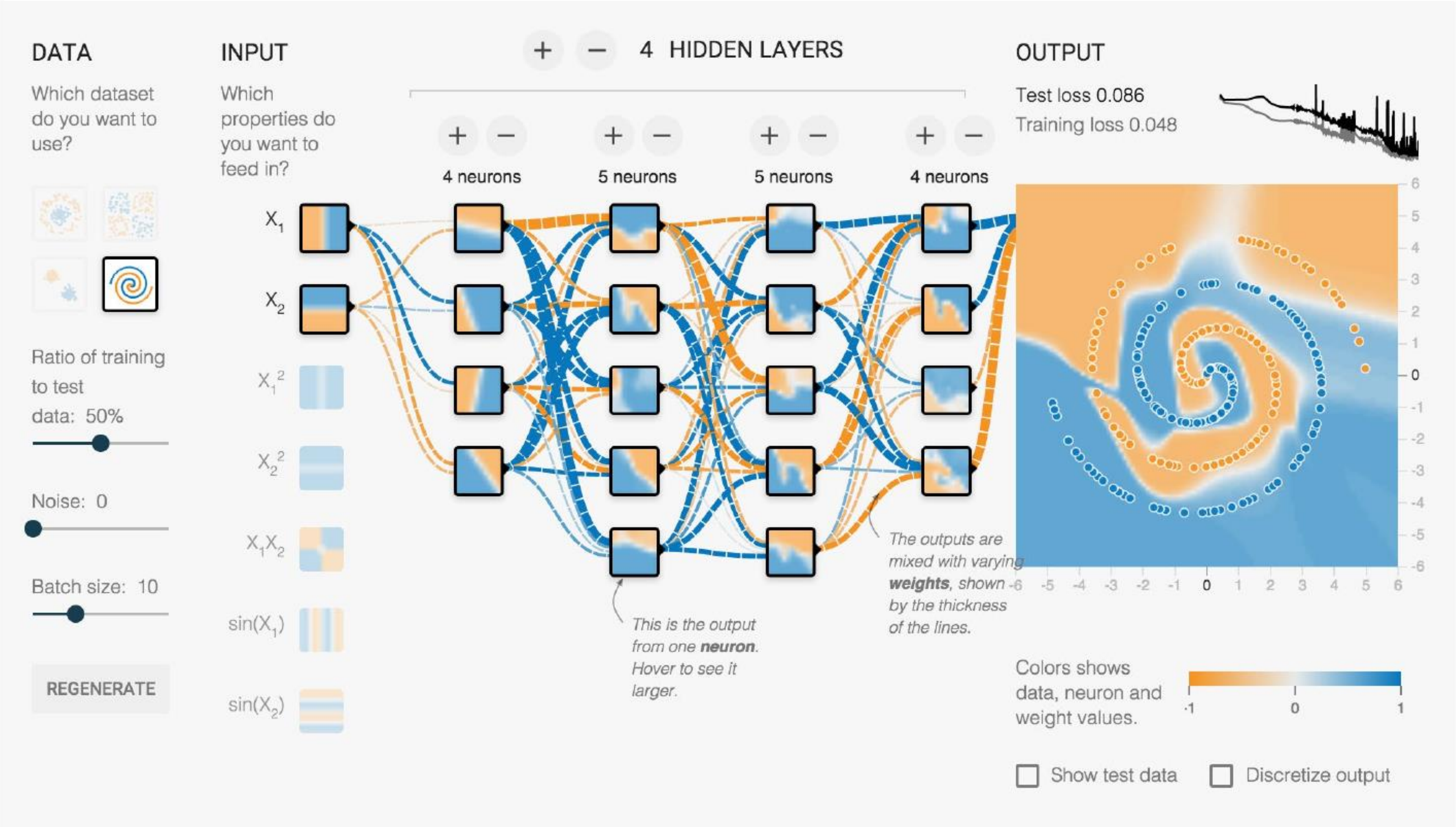
Application of Visualization: Urban Science




Components



Application of Visualization: Machine Intelligence



Academic Skills to Develop

- Maths & engineering (proficient Python)
- Reading
- Literature survey and academic writing
- Presentation

What is version control?

Managing multiple versions of documents, programs, web sites, etc

Version control is your friend – Why?

- For working by yourself:
- Gives you a “time machine” for going back to earlier versions
- Gives you great support for different versions (standalone, web app, etc.) of the same basic project
- For working with others:
- Greatly simplifies concurrent work, merging changes
- Basic skill for internships/industrial job

Git and Github

- 1. cd to the project directory you want to use
- 2. Type in git init
- ○ This creates the repository (a directory named .git)
- ○ You seldom (if ever) need to look inside this directory
- 3. Type in git add .
- ○ This adds all your current files to the repository
- ○ Period means “this directory”
- 4. Type in git commit –m "Initial commit”
- 5. type in git push origin master

How to version control: git and github

- git clone *URL*
- git clone *URL mypath*
- These make an exact copy of the repository at the given URL
- git clone git://github.com/*rest_of_path*/*file.git*
- Github is the most popular (free) public repository
- All repositories are equal